

Instructions for the reading week project:

There are two forms of assessment based on group work for ST211. The first is the Reading week mini project which is worth 10% of the final grade and the second is the end of year project group component which is worth 55%. There is also an individual group component in the end of term project which is worth 35%.

The reasons for having the Reading week mini project are as follows:

1. Many students with quantitative backgrounds do not have a lot of experience in report writing and do not know what a technical report of a statistical analysis should contain. As a consequence, there is a lot of anxiety around the end of year report. Students typically pick up the technical aspects and understand how to run regressions, however while numerical results are correct, they are often poorly explained and interpreted. The Reading week project is a guide to report writing and by providing an ideal report, we prepare the students for the longer and more important end of year report.
2. It makes sense to have a trial run under pressure of the group dynamics. You have to sign a group work commitment statement (on the RW Moodle page) and at the end you have to declare how much you contributed. So you should pull your weight and work together with your group mates. It is often the case that a group that cannot collaborate during this project has trouble in the end of term project too. Please be honest and let me know if this is the case for your group.
3. Each group gets an in-depth feedback session with me to discuss the project. This is the main feedback for this course.
4. **NOTE:** The RW project is marked in multiples of 5, no-one fails (unless they do not hand in anything) and very few people get a first. The reason is that the main aims of the RW project are to prepare you for the final projects and to get feedback. The overall value of the project is low enough that the grade you get will be very unlikely to determine your final mark in the module but high enough to get you to engage.

I expect you to fully engage in this exercise even if it is worth a small proportion of your final grade. My hope is that you take advantage of it as a learning tool which will help you improve on your end of year project and will be useful to you in the future should you need to write reports.

Explore your data, perform a regression analysis. Write up your findings. You only know how to do a few things and I do not expect you to do more than that. NO extra points will be awarded for doing things we have not covered in class.

Group work

I expect you to work as a group. During the feedback session, I will ask groups with people who contributed little why this was the case and how to improve. In most situations when this happens it is everyone in the group's responsibility. Therefore I will speak to any student who is deemed to have contributed substantially less than their group mates and I will look very negatively on students who treat group members unfairly, whether by falsely reporting that they did not contribute, by excluding group members and making it difficult for them to contribute. If you are unwell or are unable to fully participate in the project in any way, please get in touch with me and your group members asap.

NB: You will be asked to produce meeting logs to document your collaboration and contribution in the end of term project.

Each group will be invited to book an appointment with me to come and talk about your feedback. I strongly recommend that you do this. Groups that have done this in the past tended to have higher grades in their projects (although that may be because people who book meetings are keener and therefore also tend to do better regardless of the meeting!)

Data

Your group works for a statistical consultancy company (you choose the name) . The company has been contacted by the Department for Education (DfE) to analyse data from a study they have commissioned. They are giving you a reduced dataset to set up the analysis before you go into a secure environment to analyse the full dataset. They want you to produce a technical report that will go to their statistical experts to for review and also an executive summary for their non-quant upper management.

This dataset contains a merged version of the Next Steps Study and the National Pupil database (NPD). The latter contains information on examination scores at various stages (Key Stage 1,2,3 and 4). The former, Next Steps (NS), previously known as the Longitudinal Study of Young People in England (LSYPE), follows the lives of around 16,000 people in England born in 1989-90. The study began in 2004 when the cohort members were aged 14, with an original sample of 15,770 people. Cohort members were surveyed annually until 2010, and the next sweep after this was when they were aged 25, in 2015-16. The data we look at are for 2005-6 and the NPD and NS have been linked so that we can see information on the Key Stage 3 (11+ exams) and the Key Stage 4 (GCSE) scores as well as information about their family background, study habits, attitude to school and an index of deprivation (IDACI_n).

The data dictionary is on the Moodle page. I have excluded some of the variables from the data set. You will have to search the data dictionary for your variables. Please post in the forum if you have questions or are struggling. R Markdown I would prefer you to write your reports in Markdown. I will, however, not penalise groups who do not do this. Templates are available on Moodle.

Aim

The main aim of the project is to understand what factors affect the Key Stage 4 exam score. The outputs of the exercise should be the following:

1. An exploratory analysis based on plots (scatter and/or boxplots)
2. A multiple linear regression model
3. A report which explains what the group has done to arrive at this model
4. An interpretation of the model for a lay audience (e.g. your non-statistician friends and family)

Preliminary exploration and Analysis

1. Load the data into Rstudio

2. Explore the data using plots. Because most of the predictors are categorical, make sure you use boxplots.
3. Decide what to do about missing values (see report part below for details). You can manipulate the data outside of R (e.g. in Excel) if you prefer (although using the subset function makes this sort of manipulation relatively easy).
4. Once you have explored the data, write up what you have discovered in a paragraph.
5. Next reduce the number of some of levels of some of your categorical variables. You can base this on the boxplots, e.g. by merging levels that are not substantially different in the boxplots. You can also base this on small numbers, e.g. if some subgroups are very small, merge these subgroups. Make sure you justify any merging (e.g. in a short list at the back of the report – this will not be included in your word count).
6. Run some regressions. Decide whether you want to remove some predictors. Bear in mind that some predictors may be highly correlated with one another (e.g. k3ma, k3sc, k3en). You could even consider combining these (again you'll need to justify this). There may be others so check for collinearity.
7. Once you have arrived at a model you like (based on diagnostics, residual plots, but also, your own understanding of the data) write it up. This means describing briefly what you did. I give some pointers below.
8. Reflect briefly on how the model you have arrived at agrees/disagrees with your initial findings based on the plots. Remember that there is no right answer here. You should not try and twist your results to agree with your initial findings or intuitions.
9. Comment on whether this is a good model – i.e., would you be happy for policy decisions to be based on this regression? Remember it does not have to be a good model and you should be honest about this. Justify your answers.
10. Interpret the regression results.
11. Once you have written up the report as a group, one group member will upload it to Moodle. There will be a portal set up to do this. Your report should be called "RW_group" followed by your group number
12. You should also upload your R code as a txt file.

Allocate your time well. For example, if one member of the group can contribute at the beginning of the week more than at the end, then maybe they can do more of the exploratory analysis and write that up. Have someone in charge of merging categorical variable levels together (that person should know how to do it!) but decide together which variables to merge. I suggest everyone do the regression part so that you can all decide what goes in the report. I appreciate that this sounds hard (and it is). It is not my intention to fail anyone (unless they do not hand something in). The aim is to challenge you so that we can have a productive feedback session that prepares you for the end of term projects. Use the forum if you have questions.

Report

For the Reading week project, you will need to explore the data using plots and run a regression analysis as detailed above. This part is about writing up your findings. I suggest that you carefully read the instructions below and try and answer all the questions. While these do not necessarily cover all the relevant aspects, they cover the most important ones. Once you have written up your answers try and link them together to form coherent arguments.

[Stats report writing](#) is a good place to look for an idea of how to structure the different parts of the report and has good pointers for how to write in the appropriate style.

Keep it simple – because a) this means you are less likely to say the wrong thing and b) this report is worth only 10% of a half module so it does not need to be perfect. It is a learning experience.

Stay factual – do not invent things! Stick to what you know. There are parts of the report that ask you to explain why you think something is happening. Stick to the most likely explanations. If you cannot think of one, then say so. You are not a subject matter expert, so I do not expect you to know.

Be clear – use simple and plain language. Using “fancy” words does not make you sound intelligent. However, do not use colloquialisms (e.g. so basically the model was rubbish) even though I may use such language in lectures.

Choose what you say carefully – Writing irrelevant stuff to increase the word count until you hit the limit is not going to earn you marks. Concise and clear reports are best. You are conveying factual information, not writing a novel.

Do Not

- Include R code output in the report. If you use Rmarkdown then make sure code is hidden.
- Go over the Stated word limit.

Writing the report

Below are more detailed instructions on what sections there should be in a report and what you should write in them. Also look at the ideal report I wrote. Note that this report includes outlier analysis which I do not expect in this report.

The report should not exceed 2000 words but if you can say everything in fewer words that would be great!

Introduction (short and to the point)

The introduction should be a brief summary of the rest of the report. It should clearly state the question we are trying to answer and why we are interested in this question in the first place. In this section you should also give a brief background on the data and why it is appropriate to use it to answer this question. You will need to do a small amount of research. Finally, you should provide the final model and list which variables affect the ks4score in words as well. Please **do not** write down the regression equation. I know that the sample report does this, but I prefer you not to.

Think about the following when completing this part:

- What is the question of interest?
- How can the data answer this question?

- How was the original data set gathered?
- What is the answer to the question of interest given your results? Write a brief summary.

Exploratory analysis (exhaustive, but doesn't have to list every variable)

In this section, you describe briefly what plots you considered and how you interpreted them. For predictors that appeared to have no effect (i.e., all the boxplot means were very close AND they were non-significant in the regression) you can simply state that they were non-significant and not further considered. This is not necessarily what you would do in the final report, but it is sufficient here. Also consider which predictors are likely to be dependent. There is no need to include the plots. I have also run the analysis and know which plots matter. You should also decide at this stage whether any of the levels of some of the categorical predictors should be merged. List this at the back of your report. It won't count towards the word count. Consider answering the following questions:

- Which predictors appear to affect the ks4score?
- Which do not?
- Which predictors do you think are likely to be associated with one another and why?
- Which should you include in the regression (this may be all of them or only one). Justify your decision.
- Use tables to summarise the information if you prefer.
- Which categorical predictor levels can/should be merged? Justify your decisions.

There are **missing** values in the data (termed missing or -9,-1). Typically, they are missing for one or more but not all the predictors. Decide what to do about them.

1. One suggestion is to reduce the data to a complete case data (so only the individuals who have data for all the predictors you think are important initially). Then, if some of those predictors turn out to be non-significant, add back the individuals for whom values for those predictors, but not the significant ones, were missing.
2. Another approach is to start with few predictors that are important and just remove the individuals who have missing values for those. Then add other predictors reducing the data set as you add them.
3. A final suggestion is, for the categorical variables, to consider them as a level of the predictor. If you choose this path, then make sure you know think about how to interpret them. It has the advantage that you have to do less data manipulation. There are more sophisticated ways of dealing with missing data, but these are beyond the scope of this course and no additional marks will be given for trying to use them (even if this is done correctly).

Analysis (this is the meat of the project, make sure it is clear)

This part describes how you got to the final model. There are typically two ways people use:

1. You can start from a model with a few predictors that you think should be important, keep those that are significant and then add all of the other predictors one by one (or in sensible groups) and you keep the significant variables. This is called *forward selection*.

2. You could start with the full model including all the predictors and remove predictors that are not statistically significant one by one (or in sensible groups). This is termed *backward elimination*

Neither of these approaches is perfect and there are many automated versions. I don't recommend using these because it defeats the point of understanding how the process works and what problems it might have. For both approaches, I will possibly consider cases where predictors have high absolute values and/or borderline significance, however these need to be justified. In any case, you should consider the other diagnostic statistics, the residual plots and your own understanding as well as the p-values. **Your goal is not to find the model that optimizes the diagnostic statistics but that does well in the diagnostics and that can be interpreted easily.**

This means you need to think first of all about what variables you think should matter and are good predictors of ks4score. You should pay special attention to categorical variables, in particular deciding on relevant baselines. Review the Week 5 videos to make sure you understand how these work and how R deals with them.

- Write down a chronological list of what you did. Make sure it follows your R code.
- Then write it up. This does not have to have any flowery language, just the basic steps. You can divide it into subsections (Initial model, final model) or just have a series of paragraphs.
- Don't invent extra steps to fill up the space. If you have little to say, say little.
- When/why did you decide to remove or add a predictor?
- All decisions will need to be explained, although if you use the same decision to remove a number of predictors you can say "we removed x1, x2 and x3 because..."
- Refer to plots and report diagnostics where relevant, however these plots do not need to be included!

Plots (preferably no plots)

The plots that are important are those that have triggered a decision to make changes to the model or show that the model is inadequate in some way. At this stage you do not yet have the knowledge to make many changes to the analysis, however if the residual plots show violations of the assumptions, then you should say so, **However, you do not need to show ANY plots** as long as you explain why a plot triggered a decision. I would prefer a short report with only the essentials to a long report that includes every plot you produced.

Results (this is a table and a brief overview of highlights)

Tables are very important in a statistical report. The tables I want to see should only contain your final model(s) with mostly significant values and should look like the table at the top of the next page. A report without a table automatically loses 5 points.

Notice that I include a non-significant predictor because it is one of the levels of Political party for which the remaining levels are significant. There is also another level of Political party which is Conservative. Why is it not displayed? **You should NOT copy and paste the R output.**

If you wanted to show the output of a number of models, you could comni

Name of predictor	Estimate	Standard dev (or 95% confidence intervals)	p-value
age	0.5	0.04 or (0.42,0.58)	0
Political party – Labour	1.2	0.1 or (1.0,1.4)	0
Political party– Lib dems	0.5	0.2 or (0.1, 0.7)	0.05
Political party – Others	0.6	0.6 or (-1.2,1.8)	0.3
Diagnostics	R ² =0.3	Res std dev 44	

Notice that I include a non-significant predictor because it is one of the levels of Political party for which the remaining levels are significant. There is also another level of Political party which is Conservative. Why is it not displayed? **You should NOT copy and paste the R output.**

If you wanted to show the output of a number of models, you could combine them as they are likely to be nested and share many predictors.

If you are using Rmarkdown, in Moodle there is a link to a website that generates RMarkdown tables for you if you fill in the values if you are using RMarkdown. You can use that or produce a Word document output and add a table to that. The output of `display()` has the right amount of information for a table, but I also want the p-values. Do not include additional output (e.g. of `summary()`). *Create a table – do not import R output.*

Very briefly list the most variables. A full explanation is given in the final section “Interpretation & Conclusions for a lay audience.” Try not to repeat the same sentence as this can make for boring reading.

Comments about the data/analysis (to the point, but important)

Most data sets are not perfect because they are based on what people have reported. Some people will have refused to answer some questions or any question. Others will have answered them incorrectly because they feel the truth reflects badly on them or because they do not remember. Your data are a subset of a larger dataset which is in turn a subset of a larger dataset. The subset I used to generate your datasets contains only individuals who responded to some of the questions. Some of the predictors still contain missing values (called “missing” in the data). Comment on how your decision in how to deal with the missing data may have affected the results and the reliability of your analysis. In this section try and answer the following questions:

- Thinking about your predictors specifically do you think your results (coefficients) would have been different if everyone in the population had responded? I.e. could you generalise the results to all young people?
- Do you think this has the potential to seriously impact on the results? Explain your answer.
- Are there any variables where individuals might have lied when answering? Why?

- Are any of your results counter-intuitive? In what way? Remember that this is not necessarily bad nor does it reflect poorly on your model. You should not try and manipulate your model so the results agree with your intuition.
- Are there any serious flaws in your analysis that you cannot fix due to lack of expertise or because the data are flawed? Explain what you mean.
- Do you think the values that are missing are missing in a systematic way? If so, what impact could this have on the analysis?
- What other flaws are there in the data?

Executive summary (max 500 words)

The conclusions and interpretation section links back to the introduction and covers in more detail the interpretation of the results. This is the most important part of the report for non-statisticians. Because it must be easy to understand for a non-statistician it is the part where you need to write most clearly and simply. **Use NO technical language.** Think about the following points when you write up this section:

- How would you explain the results to your sister/brother/mother/father/friend who is not a statistician? The upper management are managers, policy makers, education experts and do not necessarily have quantitative backgrounds.
- One way to think about this is to look at 3-4 different types of people from your data set and calculate the expected ks4score for them. This can make things easier to understand. For example say “a male from a single parent household and five GCSEs is expected to get ks4score of 50. On the other hand a girl with the same factors is expected to get 55”. Highlighting this difference in the outcome via an example really helps. A few cleverly chosen examples can be very illuminating.
- Make sure this part flows. Try it on your sister/brother/mother/father/friend and see if they understand it. Simple is best. You can use a table or charts as well to show the results, but choose these wisely, a complex plot needs to be explained in the text.
- Based on this data set what would your recommendations be (if any) to a government representative, bearing in mind that this is a subset of the actual dataset.
- Do you think the final model is good enough to be used for decision making? It is important to say that the model is not particularly good if it isn't particularly good (e.g. residual plots indicate non-normality, diagnostics are poor).
- If you have merged levels in a categorical variable, or combined information across scores, do you need to report this?

The ONS produce regular bulletins on the analyses they do, it is worth having a look how they convey information to non-statisticians. The equivalent of the executive summary is usually at the beginning of their reports and is made up of 4/5 bullet points. You can adopt this style if you like. They often use

plots, but they are well thought through and often more relevant to descriptive statistics and not regressions.